

| Manufacturer / Hersteller                                                                                                                                                                                                                  |       |                                                      | Customer / Kunde                                                                                      |                   |      | Business Transaction / Geschäftsvorgang                                                                                              |         |                                     |                         |         |  |          |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------------|-------------------------|---------|--|----------|--|
| ACME Metal Works GmbH<br>Industrial Park 123<br>52066 Aachen<br>DE<br>quality@acme-metal.example.com                                                                                                                                       |       |                                                      | European Research Institute<br>Science Boulevard 42<br>75015 Paris<br>FR<br>materials@eri.example.org |                   |      | Order / Bestellung<br>PO-23456 · 1000 kg<br>Pos. 1 · 2025-05-01<br>Delivery / Lieferung<br>DN-65432 · 1000 kg<br>Pos. 1 · 2025-05-16 |         |                                     |                         |         |  |          |  |
| Product / Produkt                                                                                                                                                                                                                          |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Product Name / Produktname                                                                                                                                                                                                                 |       |                                                      | Titanium Alloy Ti-6Al-4V ELI                                                                          |                   |      | Number / Nummer (UNS)                                                                                                                |         | R56401                              |                         |         |  |          |  |
| Batch ID / Chargen-ID                                                                                                                                                                                                                      |       |                                                      | T-65432-01                                                                                            |                   |      | Shape / Form                                                                                                                         |         | RoundBar - D 30 mm - Length 3000 mm |                         |         |  |          |  |
| Surface Condition / Oberflächenzustand                                                                                                                                                                                                     |       |                                                      | Machined                                                                                              |                   |      | Product Norm / Produktnorm                                                                                                           |         | ASTM F136 (2022)                    |                         |         |  |          |  |
| Weight / Gewicht                                                                                                                                                                                                                           |       |                                                      | 1000 kg                                                                                               |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Production Date / Produktionsdatum                                                                                                                                                                                                         |       |                                                      | 2025-05-15                                                                                            |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Country of Origin / Ursprungsland                                                                                                                                                                                                          |       |                                                      | DE                                                                                                    |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Chemical Analysis / Chemische Analyse                                                                                                                                                                                                      |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Heat Number / Schmelznummer <b>T-65432</b> - Melting Process / Schmelzverfahren <b>VAR</b> - Casting Method / Gießverfahren <b>VacuumCasting</b> - Casting Date / Gießdatum <b>2025-05-14</b> - Sample Location / Probenort <b>Product</b> |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Symbol                                                                                                                                                                                                                                     | Ti    | Al                                                   | V                                                                                                     | Fe                | O    | C                                                                                                                                    | N       | H                                   |                         |         |  |          |  |
| Unit / Einheit                                                                                                                                                                                                                             | %     | %                                                    | %                                                                                                     | %                 | %    | %                                                                                                                                    | %       | ppm                                 |                         |         |  |          |  |
| Min                                                                                                                                                                                                                                        |       | 5.50                                                 | 3.50                                                                                                  |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Max                                                                                                                                                                                                                                        |       | 6.50                                                 | 4.50                                                                                                  | 0.25              | 0.13 | 0.080                                                                                                                                | 0.050   | 120                                 |                         |         |  |          |  |
| Actual / Ist-Wert                                                                                                                                                                                                                          | 89.32 | 6.02 ± 0.05                                          | 3.95 ± 0.03                                                                                           | 0.18              | 0.11 | 0.026                                                                                                                                | 0.012   | 35                                  |                         |         |  |          |  |
| Mechanical Properties / Mechanische Eigenschaften                                                                                                                                                                                          |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Property / Eigenschaft                                                                                                                                                                                                                     |       |                                                      |                                                                                                       | Actual / Ist-Wert |      | Minimum                                                                                                                              | Maximum | Method / Methode                    | Status                  |         |  |          |  |
| Fatigue Test                                                                                                                                                                                                                               |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         | ASTM E466                           | ✓                       |         |  |          |  |
| Room temperature, R = 0.1                                                                                                                                                                                                                  |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Cycles (N)                                                                                                                                                                                                                                 |       | 10000                                                |                                                                                                       | 50000             |      | 100000                                                                                                                               |         | 500000                              |                         | 1000000 |  | 10000000 |  |
| Value / Wert                                                                                                                                                                                                                               |       | 650                                                  |                                                                                                       | 635               |      | 610                                                                                                                                  |         | 590                                 |                         | 570     |  | 550      |  |
| Notch Sensitivity                                                                                                                                                                                                                          |       |                                                      |                                                                                                       | 0.85 - 0.92       |      |                                                                                                                                      |         |                                     | Internal Method TS-5432 | ✓       |  |          |  |
| Physical Properties / Physikalische Eigenschaften                                                                                                                                                                                          |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Property / Eigenschaft                                                                                                                                                                                                                     |       |                                                      |                                                                                                       | Actual / Ist-Wert |      | Target/Min / Soll/ Min                                                                                                               |         | Maximum                             | Method / Methode        | Status  |  |          |  |
| Density                                                                                                                                                                                                                                    |       |                                                      |                                                                                                       | 4.43 ± 0.01 g/cm³ |      | 4.43 g/cm³                                                                                                                           |         |                                     | ASTM B311               | ✓       |  |          |  |
| Supplementary Tests / Ergänzende Tests                                                                                                                                                                                                     |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Test / Prüfung                                                                                                                                                                                                                             |       | Result / limits / Ergebnis mit Grenzwerten           |                                                                                                       |                   |      |                                                                                                                                      |         | Method / Methode                    |                         | Status  |  |          |  |
| Microstructure Examination                                                                                                                                                                                                                 |       | Equiaxed alpha with intergranular beta               |                                                                                                       |                   |      |                                                                                                                                      |         | ASTM E407                           |                         | ✓       |  |          |  |
| Ultrasonic Inspection                                                                                                                                                                                                                      |       | Yes - No indications greater than reference standard |                                                                                                       |                   |      |                                                                                                                                      |         | ASTM E2375                          |                         | ✓       |  |          |  |
| Surface Quality Assessment                                                                                                                                                                                                                 |       | Class 1 - Medical Grade                              |                                                                                                       |                   |      |                                                                                                                                      |         | Visual Inspection per ASTM F136     |                         | ✓       |  |          |  |
| Alpha Case Depth                                                                                                                                                                                                                           |       | 5 µm - max 25 µm                                     |                                                                                                       |                   |      |                                                                                                                                      |         | Microhardness Traverse              |                         | ✓       |  |          |  |
| Grain Size Distribution                                                                                                                                                                                                                    |       | 8 - 10 ASTM No. - target 7 - 12 ASTM No.             |                                                                                                       |                   |      |                                                                                                                                      |         | ASTM E112                           |                         | ✓       |  |          |  |
| Hardness Profile (ASTM E384), values in HV                                                                                                                                                                                                 |       |                                                      |                                                                                                       |                   |      |                                                                                                                                      |         |                                     |                         |         |  |          |  |
| Distance from surface (mm)                                                                                                                                                                                                                 |       | 0.1                                                  | 0.5                                                                                                   | 1                 | 2    | 3                                                                                                                                    | 5       | 10                                  | 15                      |         |  |          |  |
| Value / Wert [HV]                                                                                                                                                                                                                          |       | 345                                                  | 350                                                                                                   | 352               | 350  | 349                                                                                                                                  | 348     | 351                                 | 347                     |         |  |          |  |



ACME METALS

# Digital Material Passport / Digitaler Materialpass

ID **DMP-METAL-005** - Version **1.0.0**  
Issue Date / Ausstellungsdatum **2025-05-17** - Certificate Type / Zertifikatstyp **EN 10204 3.1**

## Validation / Validierung

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.

Validated by / Validiert von **Dr. Markus Weber**, Head of Metallurgy, Research & Quality - 2025-05-17